

Time-independent Schrodinger equation

Let $H(x)$ be a time-independent Hamiltonian. Let $\psi(x)$ be a function such that $H(x)\psi(x)$ has eigenvalue E . Hence the time-independent Schrodinger equation

$$H(x)\psi(x) = E\psi(x)$$

Recall time-evolution operator $U(t, t_0)$.

$$U(t, t_0) = \exp\left(-\frac{i}{\hbar}(t - t_0)H(x)\right)$$

Let $\Psi(x, t)$ be the function

$$\Psi(x, t) = U(t, 0)\psi(x) = \exp\left(-\frac{i}{\hbar}tH(x)\right)\psi(x)$$

By hypothesis

$$\Psi(x, t) = \exp\left(-\frac{i}{\hbar}Et\right)\psi(x)$$

By linearity of $H(x)$

$$H(x)\Psi(x, t) = \exp\left(-\frac{i}{\hbar}Et\right)H(x)\psi(x) = \exp\left(-\frac{i}{\hbar}Et\right)E\psi(x) \quad (1)$$

Evaluate the Schrodinger time derivative.

$$i\hbar\frac{\partial}{\partial t}\Psi(x, t) = E\exp\left(-\frac{i}{\hbar}Et\right)\psi(x)$$

Hence by (1)

$$i\hbar\frac{\partial}{\partial t}\Psi(x, t) = H(x)\Psi(x, t)$$